

DS	Session	Arrays	Question Information	Level 1	Challenge 25
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Problem Description:

For some reason, your school's football team has chosen to spell out the numbers on their jerseys instead of using the usual digits. Being great fans, you're going to be ready to cheer for your favorite players by bringing letter cards so you can spell out their number. Each fan has different favorites, so they each need to bring different sets of letters.

The English spellings for the numbers 0 to 12 are:

ZERO ONE TWO THREE FOUR FIVE SIX
SEVEN EIGHT NINE TEN ELEVEN TWELVE

Input Format:

Read a set of integers from 0 to 12, separated by spaces, representing one fan's favorite players. The last integer will be 999, marking the end of the line.

Output Format:

Print the same numbers, then a period and a space. Then, in alphabetical order, print all the letters the fan needs to be able to spell any one of the jersey numbers provided

▼ Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN) 2 3 0999	INPUT (STDIN) 8 14 0999

EXPECTED OUTPUT

2 3 0999. E E H O R T T W

EXPECTED OUTPUT

8 14 0999. E G H I T

▼ Mandatory Test Cases

Test Case 1

KEYWORD

```
char nums[13][256]
```

Test Case 2

KEYWORD

```
for (n=0;n<26;n++)
```

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

Code:

Test Case 2

TOKEN COUNT

270

Test Case 3

NLOC 40

```

1  #include <stdio.h>
2
3  int main()
4  {
5      char *words[] = {
6          "ZERO",
7          "ONE",
8          "TWO",
9          "THREE",
10         "FOUR",
11         "FIVE",
12         "SIX",
13         "SEVEN",
14         "EIGHT",
15         "NINE",
16         "TEN",
17         "ELEVEN",
18         "TWELVE"
19     };
20
21     int num;
22     int used[26] = {0};
23     int first = 1;
24
25     /* Read numbers until 999 */
26     while (scanf("%d", &num) == 1 && num != 999)
27     {
28         /* Print the input numbers */
29         if (!first)
30         {
31             printf(" ");
32         }
33
34         printf("%d", num);
35         first = 0;
36     }

```

```
36
37      /* Get the word for the number */
38      char *word = words[num];
39
40      /* Mark each letter as used */
41      for (int i = 0; word[i] != '\0'; i++)
42      {
43          used[word[i] - 'A'] = 1;
44      }
45
46
47      printf(". ");
48
49      /* Print letters in alphabetical order */
50      for (int i = 0; i < 26; i++)
51      {
52          if (used[i] == 1)
53          {
54              printf("%c", 'A' + i);
55          }
56      }
57
58      printf("\n");
59
60      return 0;
61 }
```

Output:

```
5 6 7 999
```

```
5 6 7. EFINSVX
```

```
-----  
Process exited after 7.95 seconds with return value 0
```

```
Press any key to continue . . . |
```